

1

The conformance of the table of page-1 and descriptions of page-2 seems to be inconsistent with each other.

- **Issuer, AuthorityKey, RootSubject, RootSubjectKeyID** — these fields have 'O' as conformance, which means OPTIONAL. And in description, it is written that: all of them are OPTIONAL for PAA Certificates. Fine, it makes sense
- The conformance of **Subject** is 'M', which means MANDATORY. However, in its description, **it is ALSO declared to be OPTIONAL for PAA Certificates!** The table cell and description seems to contradict each other
- The conformance of both GrantApprovals & GrantRejects is 'M'. However, from both of their descriptions: they SHALL be set only for a PAA Certificate. **So, they will not be set for other certificates e.g. PAI, DAC? Why is the conformance of these two fields 'M' then?**

=====

=====

=====

2

Page-3

Very beginning of page-3:

Any client that discovers a Anchor Node with DNS-SD and connects to the Node via CASE **SHALL check if the Anchor CAT or the Anchor/Datastore CAT is present in the NOC of the Node and the NOC chains up to the Anchor ICAC.**

Page-4

Step-2:

Once ECM or BCM is over, Ecosystem B Administrator **SHALL check that the NOC belonging to Ecosystem A Administrator (responder NOC field of the sigma-2-tbsdata) contains either an Anchor CAT or an Anchor/Datastore CAT.**

>> No check of "the NOC chains up to the Anchor ICAC."

Page-5

Very beginning of page-3:

The picture does not mention the chaining up check either (in the rectangle box in between step 8 and 9)

Page-7

Step 5b:

“check that the NOC used by the new Datastore provider (Administrator B) during the first CASE session contains an Anchor/Datastore CAT and that the NOC chains up to the Anchor ICAC.”

>> Here, both of the checks are covered.

Analysis:

- Why have they skipped the check of "the NOC chains up to the Anchor ICAC." in page-4 and page-5? Is it intentional or by mistake?
- Intentional or not, what are the issues that can be produced, at least theoretically, because of this?
- What is implemented in the code?

=====

=====

=====

3

Page-5

LsfRevision => This field SHALL be provided “if and only if” when LsfUrl is provided.

EnhancedSetupFlowTCUrl => This field SHALL be present “if and only if” the

EnhancedSetupFlowOptions field has bit 0 set.

Page-4

EnhancedSetupFlowTCRevision => This field SHALL be present “if and only if” the

EnhancedSetupFlowOptions field has bit 0 set.

EnhancedSetupFlowTCDigest => This field SHALL be present “if and only if” the

EnhancedSetupFlowOptions field has bit 0 set.

EnhancedSetupFlowTCFileSize => This field SHALL be provided “if and only if” the

EnhancedSetupFlowTCUrl field is present.

EnhancedSetupFlowMaintenanceUrl => This field SHALL be present “if” the EnhancedSetupFlowOptions field has bit 0 set.

Analysis:

So, only for EnhancedSetupFlowMaintenanceUrl, “if” has been used, instead of “if and only if”. What this means by “if p, then q” implication formula, let’s understand that with an example

- For EnhancedSetupFlowTCDigest: If EnhancedSetupFlowOptions field has bit 0 set, then EnhancedSetupFlowTCDigest will not be present. Otherwise, it must be absent (because of “if and only if”)

- For EnhancedSetupFlowTCDigest: If EnhancedSetupFlowOptions field has bit 0 set, then EnhancedSetupFlowMaintenanceUrl will not be present. The interesting thing is that it does not need to be absent if the opposite happens. In other words, in the case where EnhancedSetupFlowOptions has bit 0 not set and EnhancedSetupFlowMaintenanceUrl

Now comes these questions

- Is there going to be an issue, at least theoretically (because of this “if p, then q” implication)?
- What is implemented in the code?
- Is it possible to map the implementation to a security vulnerability/threat?
- It’s not strictly an inconsistency, unless there is a practical impact. In other words, if they meant it to be a “don’t care” case when bit 0 is not set, then it’s probably a dead end.

=====

=====

=====

4

Page-5

Operational NodeID range => 0x0000_0000_0000_0001 to 0xFFFF_FFEF_FFFF_FFFF

Page-6

Any newly-allocated Node ID SHALL be greater than 0x0000_0000_0000_0000, but less than 0xFFFF_FFEF_FFFF_FFFF, representing a value within the Operational NodeID range

Analysis:

- Why have they excluded 0xFFFF_FFEF_FFFF_FFFF from the allowed values of Joint Fabric Node ID when it is clearly within the Operational NodeID range?
- Is there going to be an issue, at least theoretically?
- Does code also exclude 0xFFFF_FFEF_FFFF_FFFF?
- It’s not strictly an inconsistency, unless there is a practical impact. In other words, if they meant it to be a “don’t care” case when bit 0 is not set, then it’s probably a dead end.